

# Evaluating Federal Funding Impact on Broadband Equity with ISP Tier-Based Analysis

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## Abstract

This project addresses the persistent “Digital Divide” in the United States by shifting from a binary analysis of internet availability to a granular evaluation of Infrastructure Quality and Funding ROI. Utilizing data from the FCC (Broadband Data Collection), US Census (ACS), and USASpending, this research classifies Internet Service Providers (ISPs) into three tiers based on national subscriber market power. The methodology introduces a Weighted Internet Provider Availability Score (WAS) at the Census Block Group level to measure competitive density. By analyzing federal funding allocations from 2020–2025, the project aims to identify “investment dead zones” where high funding has not translated into improved service tiers. The anticipated outcome is a diagnostic framework that provides data-driven policy recommendations for future broadband subsidy distribution.

## I. INTRODUCTION/BACKGROUND

This project extends my Practicum I research, which established the socioeconomic predictors of the “Broadband Adoption Gap” at the Census Block Group level [1]. While Practicum I identified where the divide exists, this phase addresses the *why* by examining the market dynamics of ISPs and the efficiency of federal interventions. A key technical hurdle in this analysis is the reconciliation of disparate datasets—federal spending and geographic service footprints—to ensure accuracy at the Census Block Group level. This research is critical for the 2026 policy landscape as the federal government prepares to allocate remaining broadband accessibility funds.

## II. PROBLEM STATEMENT

Despite billions in federal investment through programs such as the Rural Digital Opportunity Fund (RDOF) [5] and the Broadband Equity, Access, and Deployment (BEAD) program [6], many high-need areas still lack high-tier (Fiber/Cable) broadband options. The core problem is that basic availability data often hides poor service quality and a lack of provider choice. This project utilizes the full data lifecycle—collecting FCC and USA Spending data, engineering a “Weighted Availability Score,” and modeling the relationship between funding and service improvement—to quantify why specific “funded” regions remain under served.

## III. RELATED WORK

The proposed methodology is informed by current research in three primary fields:

- **Broadband Plan Disparities:** Empirical analysis of service offerings from major US ISPs reveals significant variations in plan availability and pricing relative to socioeconomic factors, providing a modern framework for decoding the digital divide [7].
- **Broadband Policy and Regulation:** Legal and economic analysis of the Broadband Equity, Access, and Deployment (BEAD) program identifies regulatory conflicts where state preemption statutes restrict municipal networks, potentially hindering federal efforts to incentivize local infrastructure development [8].
- **Policy Evaluation:** Case studies on previous federal interventions, such as the FCC’s Rural Digital Opportunity Fund (RDOF) [5] and the ongoing Broadband Equity, Access, and Deployment (BEAD) program [6], specifically focusing on their success rates in increasing regional speeds.

#### IV. METHODOLOGY/APPROACH

This research uses observational data to identify cause-and-effect relationships.

- **Data Sources:**

- FCC National Broadband Map (Fabric): Utilized for granular, location-level broadband availability data [2]. This data is as of December 2024.
- Census ACS 5-Year Estimates: Provides the socioeconomic variables necessary to study demographic trends at the Block Group level [4]. This data is as of December 2023.
- FCC RDOF Allocation data: Serves as the primary source for federal funding data, specifically tracking RDOF allocations [5]. This data is as of October 2020.

- **Technical Stack:** Python (Pandas/GeoPandas for spatial joins), Scikit-learn for modeling ROI trends and Tableau Public for creating an interactive dashboard.
- Random Forest Regression to determine the feature importance of Tier 1, 2, and 3 competitors in driving funding efficiency, supplemented by K-Means clustering to identify disparities in service levels.
- Integrated these insights into an interactive Tableau dashboard to validate the Funding Efficiency Index (FEI) through real-time policy recommendations.

#### V. DATA DESCRIPTION

- **Grain:** Census Block Group (approx. 242,000 observations). This is a balance between about 8 million censusblocks and 50 states.
- **Ethics:** This Practicum Project will Adhere to responsible research practices by using anonymized, publicly available datasets.

#### VI. DATA PREPROCESSING

- FCC Broadband data: Procured from the National Broadband Map database [9]. Two files per state (broadband and Fiber) for 50 states and DC totaling 102 files were downloaded. The Non continental states such as Alaska and Hawaii were excluded. Then all the files were appended together to a massive panda's data frame that had 178 million rows.
  - Each row was at censusblock level and had a competitor. The competitors were classified in to tiers based on provided max download speeds (1000+ mbps = Tier1, 100+ mbps = Tier2, less than 100 mbps = Tier3)
  - Then the rows were consolidated at a block group level, with counts of tier 1, 2 and 3 competitors and max download speed available at that censusblock. This resulted in a data frame with ~ 240,000 rows.
- Census data: The Census block group level data was downloaded from the U.S. Census Bureau database [4]. Using API calls, median income and population, minority population were downloaded. Then the files were combined. Minority percentage was calculated from total population and non-white population. The final data at censusblockgroup level had ~ 240,000 rows.
- RDOF Allocation data: funding data at censusblockgroup level was obtained from the FCC Auction 904 portal [5]. This was one file with ~ 62,000 rows.
- After Data preprocessing, all the data were at the same grain of censusblock group level. they were combined to a single data frame using the common field geoid – the unique 12-digit identifier of US censusblockgroups.

#### VII. WEIGHTED INTERNET AVAILABILITY SCORE (WAS) CALCULATION

- Once FCC broadband data was summarized at a censusblockgroup level, a Normalized Internet availability score had to be calculated, so that the internet score could be compared across different regions. This new feature was labeled **Weighted Internet Availability Score (WAS)**

- A weightage of 5, 3 and 1 was provided for Tier 1, 2 and 3 competitors.

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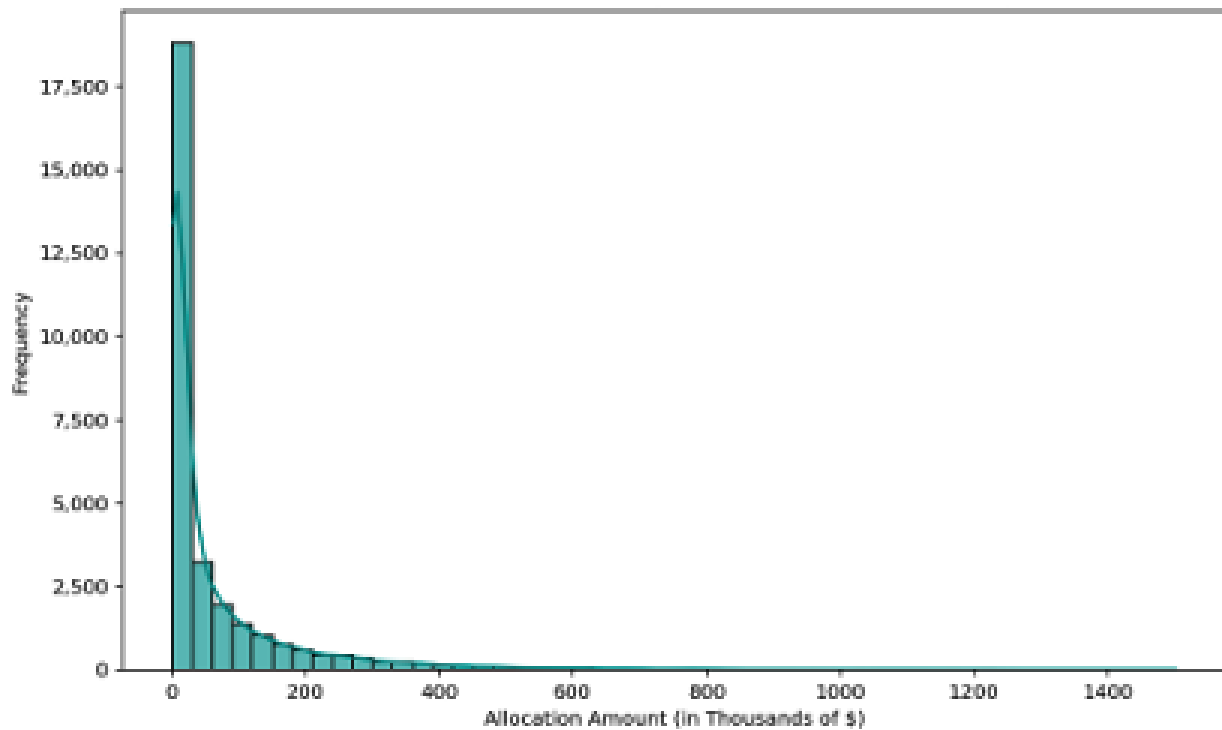
$$\text{WAS} = (\text{Tier1\_Competitor\_count} \times 5) + (\text{Tier2\_Competitor\_count} \times 3) + (\text{Tier3\_Competitor\_count} \times 1)$$

- Then, normalization was done so that the final WAS range was between 0 and 1

### VIII. EDA

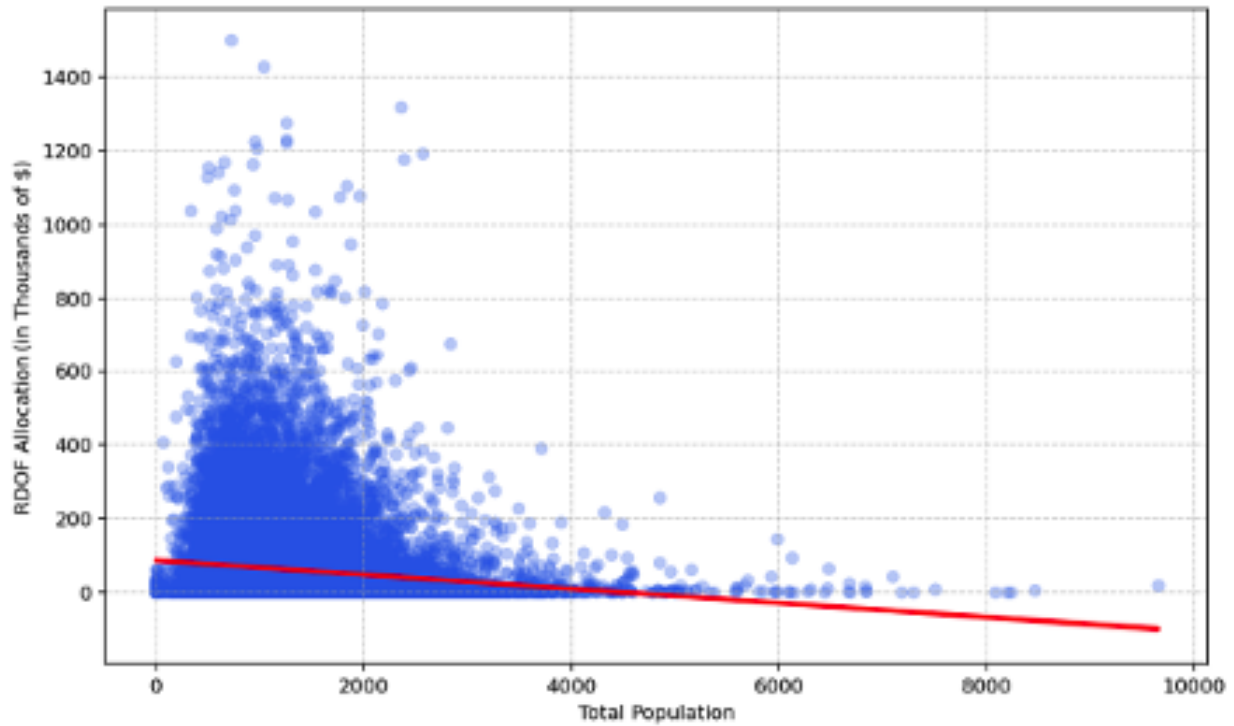
- After the data from various sources were combined to a panda's data frame, Preliminary EDA was done to find the distribution of Funds by funded amount, Population and median income.

Distribution of RDOF Funding Amounts (Funded Areas Only)



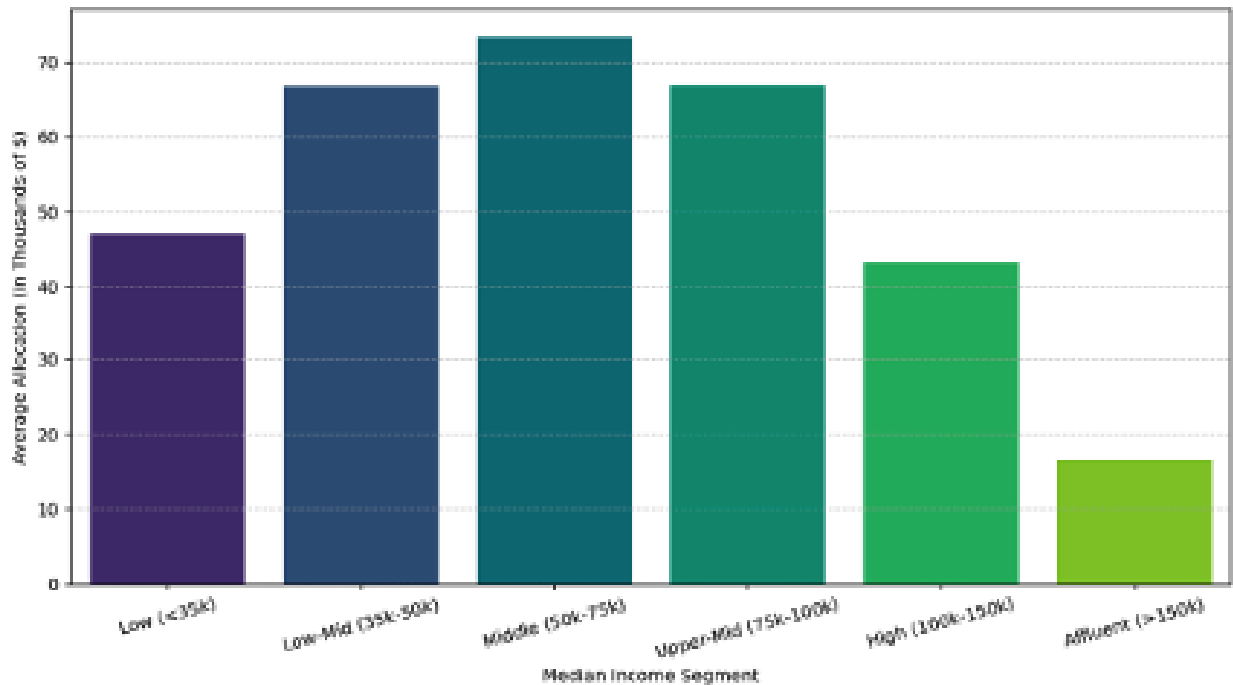
- This Histogram shows 80% funding received by was \$200,000 or less. However, some underserved areas received up to \$1.5 million.

Zoomed View: RDOF Funding vs. Low-Density Populations (&lt; 10k)



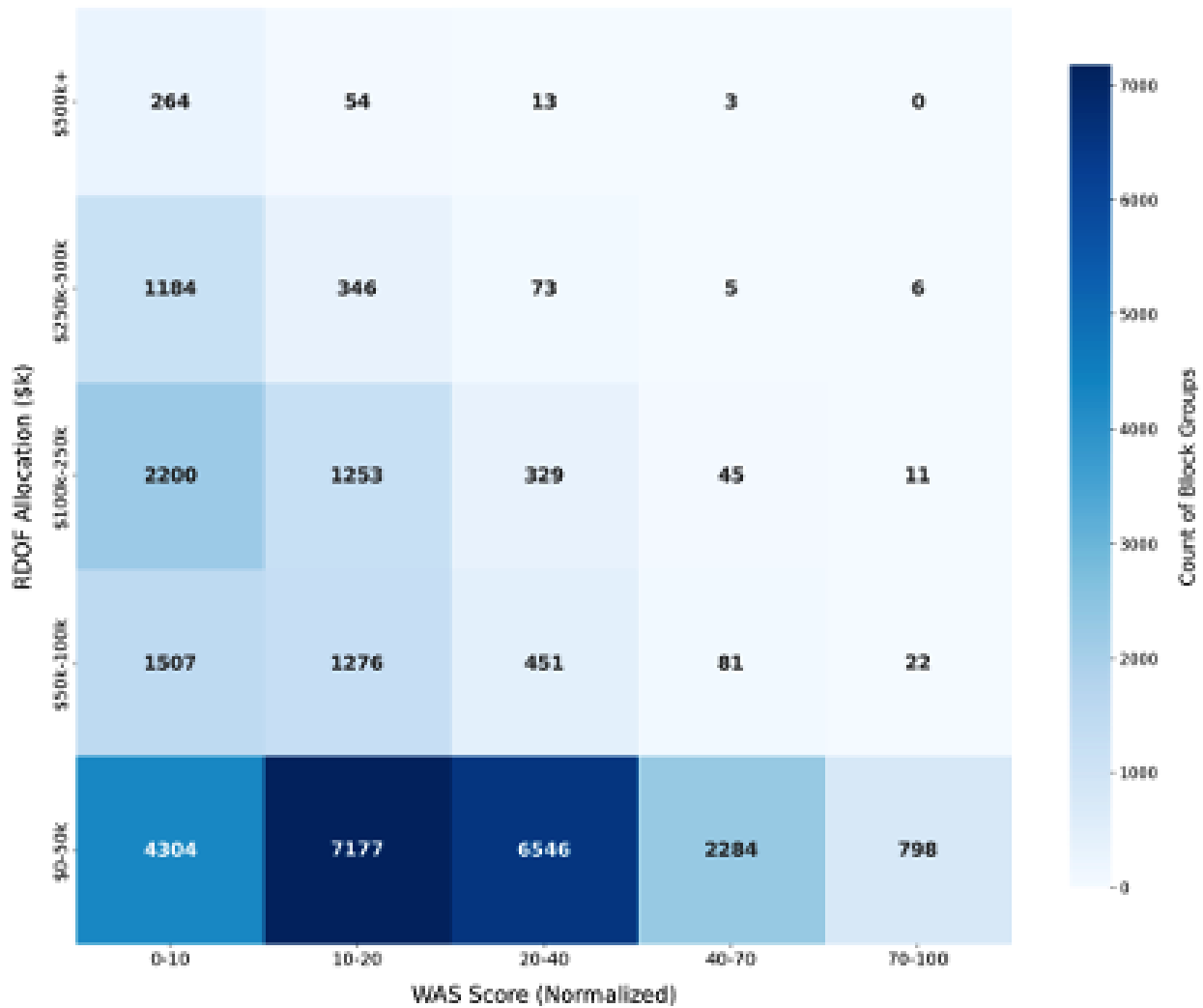
- This scatter plot shows censusblockgroups with population between 500 and 2000 received most funding. As the population increased leading to urban areas, the funding declined.

Average RDOF Allocation by Median Household Income Tier



- Plotting the funding by Median household income revealed middle class (~50K to 75K) neighborhoods received most funding, indicating a funding tilt towards sub-urban and rural areas.

Intervention Matrix: WAS (0-100) vs. RDOF Funding



- The intervention matrix highlights the relationship between the Weighted Internet Availability Score (WAS) and the RDOF allocation. A high concentration of block groups is observed in the lower WAS ranges (0-40) with funding under \$50k, suggesting that while many areas have low availability, the per-unit funding remains conservative for a large majority of them.

#### IX. FUNDING EFFICIENCY INDEX (FEI) CALCULATION

- Funding Efficiency is Points of the WAS score per \$1000 of funding.
  - $FEI = WAS / RDOF \text{ Allocation} / 1000$

#### X. SUPERVISED MACHINE LEARNING: RANDOM FOREST REGRESSION

- A Random Forest Regression was done to predict the Target – FEI.
- The Features used were:
  - Tier1, 2 and 3 Competitor counts
  - Max download speed available
  - Median Income
  - Population
  - Minority Percentage

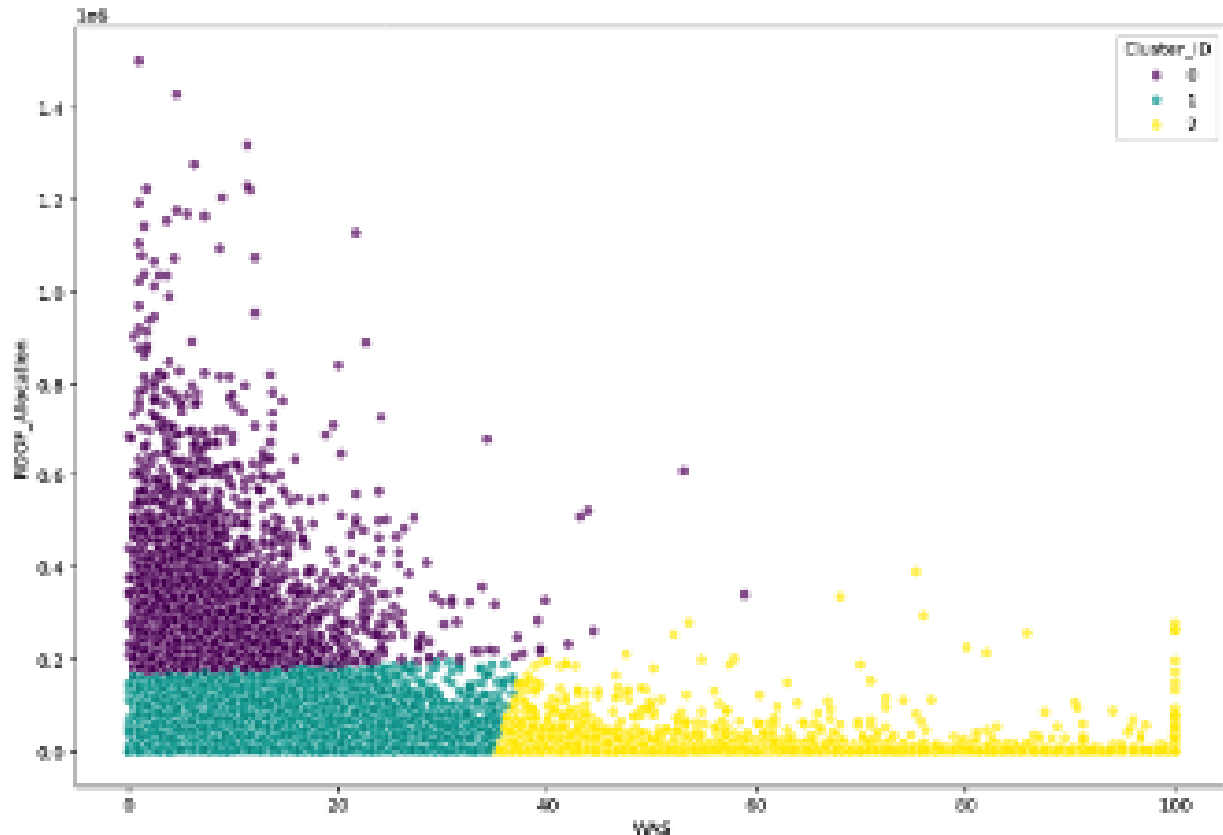
- The Hyper Parameters used were:
  - `n_estimators = 100`, `random_state = 42`, `n_jobs = -1`, `train size = 20%`
- Results of Random Forest Regression:
  - R-Squared score = 0.5013
  - Feature Importance:
    - \* Tier1 Competitors = 56.5%
    - \* Median income = 10.41%
    - \* Minority percentage = 10.07%
    - \* Population = 9.95%
    - \* Tier2 Competitors = 7.76%
    - \* Max Down Speed = 4.04%
    - \* Tier3 Competitors = 1.27%
- Interpretation:
  - An R sq of 0.5013 means the model explains about 50% of the variance in funding efficiency.
  - Tier1 competitor count is the major influencer. This is possible because the big-name competitors bid for the funds and use them to improve internet infrastructure
  - Between income, population and minority percentage, income is the biggest influencer, corroborating the EDA that was done for income distribution of funding received.

#### XI. UNSUPERVISED MACHINE LEARNING: K-MEANS CLUSTERING

- A K-means clustering was done to see relationship between the funding and resulting Internet availability score.
- Cluster Features:
  - WAS
  - RDOF funding amount
- The Hyper Parameters used were:
  - `n_clusters = 3`, `random_state = 42`, `n_init = 10`
- Results of K-means clustering (mean values)

| Cluster | WAS      | RDOF_Allocation |
|---------|----------|-----------------|
| 0       | 0.336229 | 320877.9722     |
| 1       | 0.677412 | 32984.01805     |
| 2       | 2.791202 | 10228.27454     |

### Unsupervised Personas: WAS vs. RDOF Allocation



- Interpretation:
  - Cluster 0 (purple dots) are the most funded. This cluster has low WAS score and shows the funds were allocated in the right geographies
  - Cluster 2 (yellow dots) were the least funded. Here again the WAS score is relatively high and so the funding amount is low
  - Cluster 1 (green dots) are a surprise. They are having low WAS score and yet received low funding. This cluster needs future research. One reason could be that these areas could be funded by other state level / municipality level funding programs (eg BEAD – Broadband Equity Access and Development program).

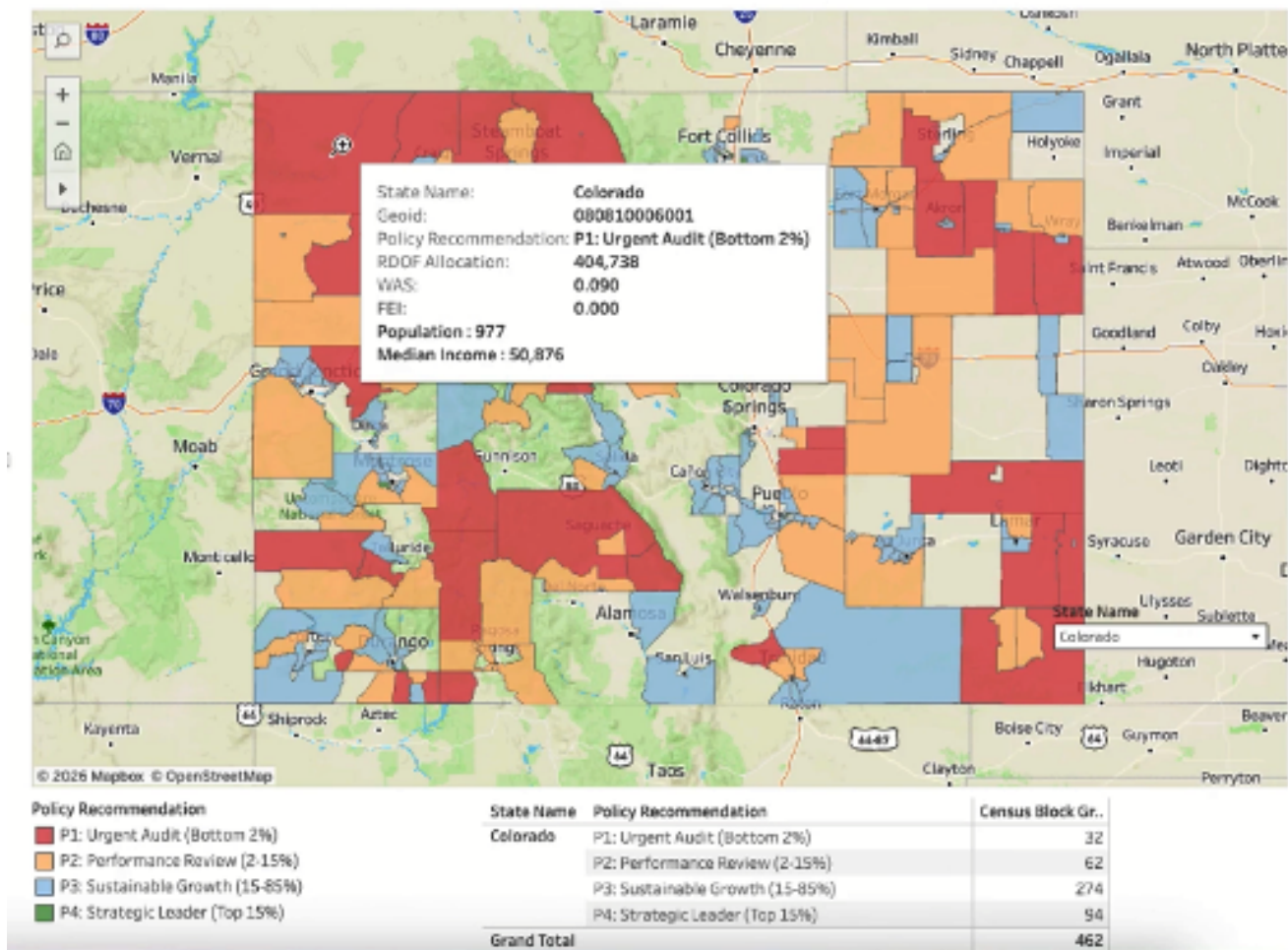
### XII. NATIONAL POLICY CLASSIFICATION CALCULATION

- A National Policy Classification / Recommendation was assigned to each censusblock group that received funding.
- RDOF amount, WAS, FEI were scaled to values of 0 to 1
- Policy\_Score was calculated as follows:
  - $\text{Policy\_Score} = \text{FEI\_scaled} * 0.4 + \text{WAS\_scaled} * 0.4 - \text{RDOF\_scaled} * 0.2$
  - RDOF scaled was given a negative multiplier so as reduce the impact of highly funded areas
- Policy Recommendation was calculated based on the percentile values of RDOF scaled.
  - Bottom 2% of Policy Score = P1: Urgent Audit
  - 2% – 15% of Policy Score = P2: Performance Review
  - 15% – 85% of Policy Score = P3: Sustainable Growth
  - Top 15% of Policy Score = P4: Strategic Leader

### XIII. INTERACTIVE TABLEAU DASHBOARD

- The features, calculated metrics and National Policy Classification was exported in a GeoJason format file.
- This File was imported into Tableau Public and plotted on a continental map of USA.
- The National Policy Calculation was color coded from Red to Green based on Priority
- A State filter was provided to zoom in on any state or compare more than one state together.
- A Table below the map provided the count of census blocks and Policy Classifications by state.
- The Tooltip was updated to show the information pertaining to the censusblock group.
- The Tableau Dashboard can be accessed at the following link: Interactive RDOF Policy Recommendations Dashboard [10].

#### Policy Recommendation based on Funding Efficiency Index (FEI)



- The map above illustrates the spatial distribution of policy recommendations across Colorado. The tooltip highlights specific metrics for a census block group, such as the Geoid, RDOF Allocation, WAS, and FEI, which collectively drive the recommendation category (e.g., P1: Urgent Audit).
- The summary table provides a breakdown of the count of census block groups for each classification, showing that the majority fall under P3: Sustainable Growth.

### XIV. CONCLUSION

- This project represents the culmination of my Master's studies by applying geospatial data science to the societal challenge of digital equity. By analyzing the interplay between ISP market tiers and



federal funding, I was able to develop a diagnostic tool that is both academically rigorous and practically necessary for bridging the broadband divide in 2026.

- This research utilizes advanced analytical techniques to quantify why specific regions remain underserved despite federal investment, providing a data-driven framework for future policy recommendations.
- Building on my professional experience as a Senior Manager in competitive intelligence within the cable and internet industry, I merged industry-specific market familiarity with the advanced data science methodologies learned in the MSDS program at Regis University.
- This project is deeply personal for me, as it will contribute positively toward solving the broadband adoption gap in the United States. Ultimately, I developed a model that transforms complex market dynamics into actionable insights, ensuring that my academic journey directly supports the goal of national digital inclusion.

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